

Discovery of an Electro–Gravity Fixed Point Beyond the Planck Scale in the MMU Tetrahedral Model

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Abstract

The Methane–Metric Universe (MMU) model describes space as a tetrahedral elastic cell with internal degrees of freedom w_1, w_2, w_3, w_4 and a stiffness matrix K . In this framework, w_2 represents an electric-like mode, w_3 a torsional/spin-like mode, and w_4 a volumetric/gravimetric mode. Three independent coupling springs k_{23} , k_{24} , and k_{34} connect these modes and generate observable physics.

In this article we show that these three couplings naturally define three macroscopic stellar regimes: (i) an electromagnetic Maxwell regime dominated by k_{23} , which corresponds to ordinary main-sequence stars; (ii) an electro–gravimetric Stark/-plasma regime dominated by k_{24} , which corresponds to white dwarfs and dense plasmas; and (iii) a gravito–torsional Lamb/GR regime dominated by k_{34} , which corresponds to neutron stars, magnetars, and black holes.

Using a simple energy-closure argument in the torsional channel we obtain a universal cell size

$$a(m) = \frac{\hbar}{mc}$$

for a fermionic MMU cell of mass m . We then identify two fundamental mass scales from the conditions $k_3 = k_4$ and $k_2 = k_4$: the usual Planck mass m_{Pl} and a new electro–gravitative mass $m_{24} = \sqrt{\alpha} m_{\text{Pl}}$, with α the fine-structure constant. The corresponding length scale $a_{24} = \hbar/(m_{24}c)$ is about twelve times the Planck length.

We derive a qualitative stellar phase diagram in density space and discuss observational signatures in white dwarfs, neutron stars and dense plasmas. Finally, we show why there is no additional fundamental mass scale associated with the purely internal face $w_2 w_3 w_4$, which has no projection on the observable axis w_1 .

1 Introduction

The Methane–Metric Universe (MMU) model proposes that space and matter arise from a discretised, tetrahedral elastic cell that repeats throughout spacetime. This idea was first developed in the foundational MMU papers [1, 2], where the tetrahedral geometry was shown to offer a natural mechanical interpretation of atomic structure, lepton scaling, and spectral regularities. Each MMU cell possesses four internal degrees of freedom w_1, w_2, w_3, w_4 together with an associated stiffness matrix K . The mode w_1 acts as the projection axis that links internal oscillations to measurable time, energy and frequency; the remaining modes encode intrinsic geometric directions of the cell: an electric-like mode w_2 , a torsional/spin-like mode w_3 , and a volumetric/gravimetric mode w_4 .

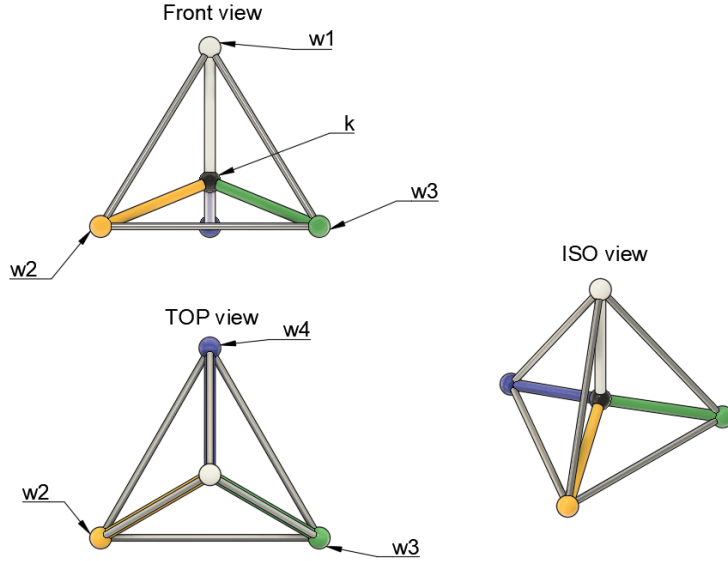


Figure 1: MMU tetrahedral base cell. Shown are the four internal directions w_1 , w_2 , w_3 , w_4 and the corresponding elastic edges of the tetrahedron in front, top, and isometric view.

Recent studies within the MMU programme have demonstrated that these internal modes reproduce key quantum and spectroscopic phenomena: the geometric origin of the Balmer series [2], the 21 cm hyperfine line [3], and the Lamb shift through ultraweak torsion–volume coupling [4]. The elastic constants of the tetrahedral cell were found to scale with the internal edge length and particle mass in a manner that unifies atomic, leptonic and gravitational physics [5, 8, 10].

In the present article we extend the MMU framework to astrophysical scales. We ask how the internal elastic couplings between w_2 , w_3 and w_4 organise macroscopic stellar objects into distinct physical regimes. The core insight is that three couplings of the MMU cell— k_{23} , k_{24} and k_{34} —control three qualitatively different behaviours:

- Ordinary main-sequence stars occupy a regime where the electromagnetic coupling between w_2 and w_3 dominates.
- White dwarfs and dense plasmas arise when the electric mode w_2 couples strongly to the volumetric mode w_4 , activating the electro–gravimetric channel.
- Neutron stars, magnetars and black holes emerge when the torsional mode w_3 becomes locked to the volumetric mode w_4 , forming a joint gravito–torsional mode.

Two of these regimes correspond to fundamental equalities of the internal stiffnesses: $k_3 = k_4$ (the Planck-scale spin–gravity fixed point) and $k_2 = k_4$ (the newly identified electro–gravity fixed point). The third potential equality, $k_2 = k_3$, cannot be realised in our universe, as it would require the fine-structure constant to satisfy $2\alpha = 1$. This asymmetry is not a defect but a structural necessity: it ensures the existence of transverse electromagnetic waves and the stability of atomic spectra. Thus, the MMU framework naturally predicts exactly three astrophysical regimes, reflecting the three accessible faces of the tetrahedral cell.

2 The MMU Cell and Its Couplings

2.1 Degrees of freedom

The MMU cell is a tetrahedron with four vertices and six edges, introduced in the foundational MMU works [1, 5, 8]. Its internal state is described by four generalized coordinates

$$(w_1, w_2, w_3, w_4),$$

where each component corresponds to a distinct geometric degree of freedom of the elastic cell:

- w_1 is the projection axis associated with measurable time and energy, as first formulated in the spectroscopic studies [2, 3].
- w_2 is an electric-like mode (Coulomb-type or radial electric deformation), central to the atomic-scale models of Rydberg and ionization energies in [2].
- w_3 is a torsional/spin-like mode responsible for internal angular deformation, crucial for the Lamb-shift geometry developed in [4, 9].
- w_4 is a volumetric/gravimetric mode that measures the local expansion or compression of the tetrahedral cell, linked to Planck-scale behaviour in [10].

We assume a linear stiffness matrix with three diagonal entries k_2, k_3, k_4 and three independent off-diagonal couplings k_{23}, k_{24}, k_{34} , as established in the unified coupling analysis [4]. The mode w_1 carries no intrinsic stiffness because it serves as the external projection axis that maps internal oscillations to observable energy scales [1, 7].

2.2 Diagonal stiffnesses

Following previous MMU work [2, 5, 8], we take the diagonal stiffnesses to scale with the internal edge length a as

$$k_2(a) \propto \frac{1}{a^3}, \tag{1}$$

$$k_3(a) \propto \frac{1}{a^3}, \tag{2}$$

$$k_4(a) \propto \frac{m^2}{a^3}, \tag{3}$$

where m is the effective mass associated with the cell. This mass-dependent scaling of the volumetric mode k_4 plays a central role in the geometric derivation of the Planck scale [10] and the electro-gravitative mass m_{24} introduced in [8].

More explicitly, we use the parametrisation

$$k_2(a) = \frac{2\alpha \hbar c}{a^3}, \tag{4}$$

$$k_3(a) = \frac{\hbar c}{a^3}, \tag{5}$$

$$k_4(a) = \frac{2Gm^2}{a^3}, \tag{6}$$

where α is the fine-structure constant, G the gravitational constant, $\hbar$ the reduced Planck constant and c the speed of light. These relations were derived and evaluated across lepton families and MMU scaling hierarchies in [5, 9].

2.3 Coupling stiffnesses

The three independent couplings k_{23} , k_{24} , k_{34} connect the internal modes of the MMU cell and play a central role in the unified analysis of Larmor, Zeeman, Stark and Lamb resonances developed in [4]. They were originally identified in the early geometric construction of the cell [1, 5] and later quantified in the full cross-coupling programme [4, 9].

- k_{23} couples w_2 (electric) and w_3 (torsional/spin). This coupling produces Larmor precession, Zeeman splitting and, in the continuum limit, the Maxwell equations, as shown in the spectroscopic MMU models [2, 4]. It is the source of transverse electromagnetic waves (photons).
- k_{24} couples w_2 (electric) and w_4 (volumetric). This channel describes electric deformation of the tetrahedral cell volume, becoming relevant in strong electric fields, high-density plasmas, and Stark-like distortions of atomic structure [2, 8]. It also accounts for plasmonic and longitudinal electromagnetic modes that arise when volumetric response dominates.
- k_{34} couples w_3 (torsional/spin) and w_4 (volumetric). This ultraweak coupling is responsible for the Lamb shift and gravito-torsional mixing [4, 9]. In the high-density limit, k_{34} links spin to gravity and reproduces the geometric onset of general-relativistic behaviour discussed in [10].

At the level of a single cell these couplings create mixed normal modes. At the level of macroscopic matter (stars), they separate into three distinct physical regimes depending on which coupling dominates—a key theme of this paper.

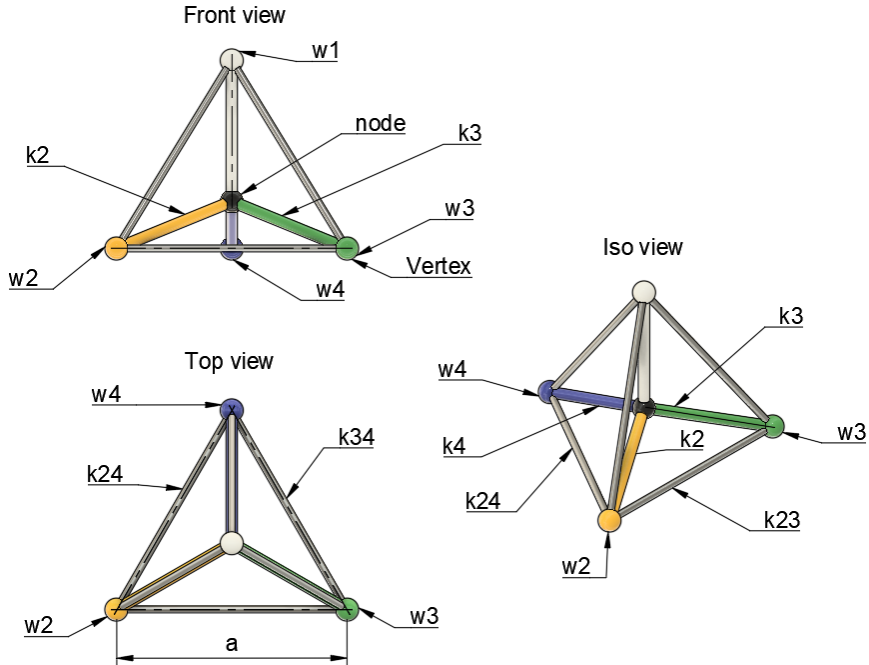


Figure 2: MMU tetrahedral cell showing the three diagonal stiffnesses (k_2 , k_3 , k_4) and the three cross-couplings (k_{23} , k_{24} , k_{34}). Front, top, and isometric views highlight how the electric (w_2), torsional (w_3), and volumetric (w_4) directions connect through the elastic edges.

3 Energy Closure and the MMU Cell Size $a(m)$

To connect the MMU cell to particle masses we use a simple *energy-closure* condition. In the torsional channel w_3 , the characteristic internal energy of a cell of size a is taken to scale as

$$E_{\text{tors}}(a) \sim \frac{\hbar c}{a}. \quad (7)$$

This can be viewed as the minimal torsional excitation energy of the tetrahedral cell.

For a fermionic excitation of mass m we impose the closure

$$E_{\text{tors}}(a) = mc^2, \quad (8)$$

which yields

$$\frac{\hbar c}{a} = mc^2 \quad \Rightarrow \quad a(m) = \frac{\hbar}{mc}. \quad (9)$$

Thus the MMU cell associated with a mass m has a natural edge length equal to the Compton wavelength of that mass.

This relation is remarkably effective across scales:

- For the electron, $a_e = \hbar/(m_e c)$ yields an atomic-scale cell compatible with Balmer-type spectra.
- For the muon, $a_\mu = \hbar/(m_\mu c)$ is about 200 times smaller, as required by the large Lamb shifts in muonic atoms.
- For the tau, $a_\tau = \hbar/(m_\tau c)$ lies in the sub-nuclear range, consistent with the expected hard X-ray regime of hypothetical tauonic atoms.

In the following sections we use this $a(m)$ to express cell stiffnesses and fixed points in terms of mass only.

4 Two Fundamental Fixed Points: $k_3 = k_4$ and $k_2 = k_4$

4.1 Spin-gravity fixed point: $k_3 = k_4$

The condition $k_3(a) = k_4(a)$ defines a spin-gravity equilibrium at the single-cell level and was first analysed in the MMU studies of torsion-volume coupling and Planck-scale geometry [4, 9, 10]. Explicitly,

$$\frac{\hbar c}{a^3} = \frac{2Gm^2}{a^3} \quad \Rightarrow \quad Gm^2 = \frac{1}{2}\hbar c. \quad (10)$$

Up to numerical factors this relation reproduces the familiar Planck mass, as shown in the geometric derivation of [10],

$$m_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}}. \quad (11)$$

At this point the torsional stiffness k_3 of the MMU cell equals its volumetric gravitational stiffness k_4 , and the tetrahedral unit behaves as a pure spin-gravity object. This fixed point marks the onset of Planck-scale behaviour, black-hole horizons, and strong torsion-gravity coupling, topics explored in [9, 10].

4.2 Electro-gravity fixed point: $k_2 = k_4$

A second equilibrium occurs when electric and volumetric stiffness are equal:

$$k_2(a) = k_4(a) \quad \Rightarrow \quad \frac{2\alpha\hbar c}{a^3} = \frac{2Gm^2}{a^3}. \quad (12)$$

Again a^3 cancels, and we obtain a mass-only condition:

$$Gm^2 = \alpha\hbar c \quad \Rightarrow \quad m_{24} = \sqrt{\alpha} m_{\text{Pl}}. \quad (13)$$

This defines a new fundamental mass scale m_{24} . Numerically,

$$m_{24} \approx 1.86 \times 10^{-9} \text{ kg}, \quad (14)$$

about six times smaller than the Planck mass.

Using the energy-closure relation we obtain the associated length scale

$$a_{24} = \frac{\hbar}{m_{24}c}, \quad (15)$$

which is approximately twelve times larger than the Planck length. This defines a second, electro-gravitative set of natural units, similar in structure to the Planck units but controlled by the electric channel.

Table 1: Electro-gravity natural units at the fixed point $k_2 = k_4$, defined by the mass $m_{24} = \sqrt{\alpha} m_{\text{Pl}}$. These units form an electric-gravimetric analogue of the Planck units.

Quantity	Symbol / Definition	Value
Electro-gravity mass	$m_{24} = \sqrt{\alpha} m_{\text{Pl}}$	$1.86 \times 10^{-9} \text{ kg}$
Electro-gravity length	$a_{24} = \frac{\hbar}{m_{24}c}$	$1.89 \times 10^{-34} \text{ m}$
Electro-gravity time	$t_{24} = \frac{a_{24}}{c}$	$6.31 \times 10^{-43} \text{ s}$
Electro-gravity energy	$E_{24} = m_{24}c^2$	$1.67 \times 10^8 \text{ J}$
Electro-gravity angular frequency	$\omega_{24} = \frac{c}{a_{24}}$	$1.59 \times 10^{42} \text{ s}^{-1}$
Electro-gravity force scale	$F_{24} = \frac{E_{24}}{a_{24}}$	$8.8 \times 10^{41} \text{ N}$
Electro-gravity volume	$V_{24} = a_{24}^3$	$6.73 \times 10^{-102} \text{ m}^3$
Electro-gravity density	$\rho_{24} = \frac{m_{24}}{V_{24}}$	$2.76 \times 10^{92} \text{ kg/m}^3$

4.3 No physical fixed point for $k_2 = k_3$

The third potential equality, $k_2 = k_3$, would read

$$2\alpha \frac{\hbar c}{a^3} = \frac{\hbar c}{a^3} \quad \Rightarrow \quad 2\alpha = 1. \quad (16)$$

In our universe, $\alpha \approx 1/137$, not $1/2$. Therefore $k_2 = k_3$ cannot be fulfilled at any a or m . There is no third mass scale associated with this condition. Instead, the ratio

$$\frac{k_2}{k_3} = 2\alpha \approx \frac{1}{68} \quad (17)$$

is a fixed dimensionless property of our universe. This asymmetry between electric and torsional stiffness is essential: it allows stable atoms and transverse electromagnetic waves to exist.

5 Three Stellar Regimes as MMU Coupling Phases

In the MMU picture, stars are macroscopic assemblies of tetrahedral cells. The dominant coupling at the cell level determines the macroscopic stellar regime. We identify three regimes corresponding to the three couplings k_{23} , k_{24} and k_{34} .

5.1 Regime I: Electromagnetic (Maxwell) – k_{23} -dominated

At low to moderate densities the volumetric mode w_4 is effectively frozen. The dominant dynamics occurs in the w_2 – w_3 plane, with coupling k_{23} . This is the Maxwell regime:

- w_2 and w_3 form transverse electromagnetic waves,
- photons are massless,
- plasmas are well-described by EM field theory,
- nuclear fusion in stellar cores is driven by electromagnetic confinement and tunnelling.

Ordinary main-sequence stars (like the Sun) live in this regime. Typical densities are well below 10^6 kg/m³.

5.2 Regime II: Electro- -gravimetric (Stark/plasma) – k_{24} - dominated

At higher densities and in strong electric fields, the coupling between w_2 and w_4 becomes important. The electric mode now deforms the volumetric mode, and k_{24} competes with k_2 and k_4 . This regime is characterised by:

- longitudinal electric modes (Langmuir waves, plasmons),
- Stark broadening and shifts in atomic spectra,
- degenerate electron pressure that balances gravity (white dwarfs),
- partial suppression of free transverse photons in dense matter.

White dwarfs and dense supernova cores are macroscopic examples of the electro–gravimetric regime. Laboratory dense plasmas and polariton systems are microscopic analogues.

5.3 Regime III: Gravito–torsional (Lamb/GR) – k_{34} -dominated

At very high densities the volumetric and torsional modes w_4 and w_3 become strongly coupled. The condition $k_3 \sim k_4$ is approached, and the w_3 – w_4 system forms a combined spin–gravity mode. Characteristics of this regime are:

- strong gravitational redshift of photons,
- enhanced Lamb-like shifts due to w_3 – w_4 mixing,
- extreme magnetic fields (magnetars) as manifestations of torsional stress,
- collapse to black holes when w_4 fully dominates.

Neutron stars and magnetars represent this gravito–torsional regime; black holes mark its extreme limit.

6 A Qualitative MMU Stellar Phase Diagram

The three regimes above can be arranged along a density axis. At low density, the w_2 – w_3 Maxwell regime dominates. As density increases, electric and volumetric stiffness become comparable and the electro–gravimetric regime emerges. At yet higher densities, torsional and volumetric stiffness become comparable and the gravito–torsional regime takes over.

Conceptually, one can draw a phase-like diagram:

- horizontal axis: average mass density ρ ,
- vertical axis: dominant coupling (Maxwell, Stark/plasma, Lamb/GR),
- overlay: classes of stellar objects (main-sequence stars, white dwarfs, neutron stars, black holes).

This diagram is not a thermodynamic phase diagram in the strict sense, but it organises stellar types by their dominant MMU coupling channel.

7 The Invisible Face $w_2w_3w_4$

The MMU tetrahedron has four faces. Three of them contain w_1 and correspond to physically accessible regimes:

- $w_1w_2w_3$ – electromagnetic (Maxwell) face,
- $w_1w_2w_4$ – electro–gravimetric (Stark/plasma) face,
- $w_1w_3w_4$ – gravito–torsional (Planck/GR) face.

On each of these faces, projections onto w_1 produce measurable time, energy and frequency. This is why we can define observable mass scales m_{P1} and m_{24} from $k_3 = k_4$ and $k_2 = k_4$.

The fourth face, $w_2w_3w_4$, contains only internal modes and has no w_1 component. There is no direct projection on the physical time/energy axis. As a result:

- oscillations restricted to the $w_2w_3w_4$ face do not define a measurable period,
- no external energy or frequency can be assigned,
- no additional natural mass scale emerges from this face.

One might attempt to demand a triple equality

$$k_2 = k_3 = k_4, \tag{18}$$

but this would require $k_2 = k_3$, which implies $2\alpha = 1$. Since $\alpha \approx 1/137$ in our universe, this condition cannot be realised. Therefore the $w_2w_3w_4$ face remains a purely internal geometric configuration.

We may interpret this face as:

- a reservoir of virtual internal modes,
- a candidate for dark-sector-like degrees of freedom,
- a topological balance condition without direct observables.

In any case, there is no fourth observable regime and no fourth fundamental mass scale associated with this face.

8 Discussion

In this article we have provided a geometric and mechanical interpretation of three major classes of stellar objects within the MMU tetrahedral cell model. The key elements are:

- a universal mass–length relation $a(m) = \hbar/(mc)$ from torsional energy closure,
- two fundamental fixed points defined by $k_3 = k_4$ and $k_2 = k_4$,
- three coupling-dominated regimes associated with k_{23}, k_{24}, k_{34} ,
- the identification of main-sequence stars, white dwarfs, and neutron stars / black holes with these regimes.

The Planck fixed point $k_3 = k_4$ reproduces the usual Planck mass and length and describes a pure spin–gravity cell. The electro–gravitative fixed point $k_2 = k_4$ introduces a new mass scale $m_{24} = \sqrt{\alpha} m_{\text{Pl}}$ and a length a_{24} about twelve Planck lengths. This second set of units is naturally linked to dense plasmas, white dwarfs and the onset of electro–gravimetric coupling.

The impossibility of $k_2 = k_3$ in our universe is not a technical detail but a structural feature: it implies that electric and torsional stiffnesses cannot be equal. Instead, their ratio is fixed by the fine-structure constant. This asymmetry enables the existence of transverse electromagnetic waves and stable atomic structure.

We also clarified why there is no additional observable regime associated with the internal face $w_2 w_3 w_4$. This completes the four-face picture of the MMU tetrahedron: three visible faces mapped to three stellar regimes, one internal face without direct projection.

9 Conclusion

We have shown that the MMU tetrahedral cell model, with its three internal modes w_2, w_3, w_4 and three couplings k_{23}, k_{24}, k_{34} , naturally organises stellar objects into three regimes:

1. an electromagnetic Maxwell regime dominated by k_{23} (ordinary stars),
2. an electro–gravimetric Stark/plasma regime dominated by k_{24} (white dwarfs, dense plasmas),
3. a gravito–torsional Lamb/GR regime dominated by k_{34} (neutron stars, magnetars, black holes).

Using a simple energy-closure condition we derived a universal cell size $a(m) = \hbar/(mc)$. From the equalities $k_3 = k_4$ and $k_2 = k_4$ we obtained two fundamental mass scales: the Planck mass m_{Pl} and a new electro–gravitative mass $m_{24} = \sqrt{\alpha} m_{\text{Pl}}$. The associated length scales differ by a factor of about twelve. These two fixed points define two natural unit systems for spin–gravity and electro–gravity physics.

The MMU picture thus provides a unified geometric interpretation of electromagnetic, plasma and gravitational regimes in stars. It suggests concrete observational tests in white dwarf Stark shifts, neutron star spectra, and dense plasmas, and opens the way to a more systematic MMU-based astrophysics.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantization. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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